Peter Fidler's method for mapping his travels

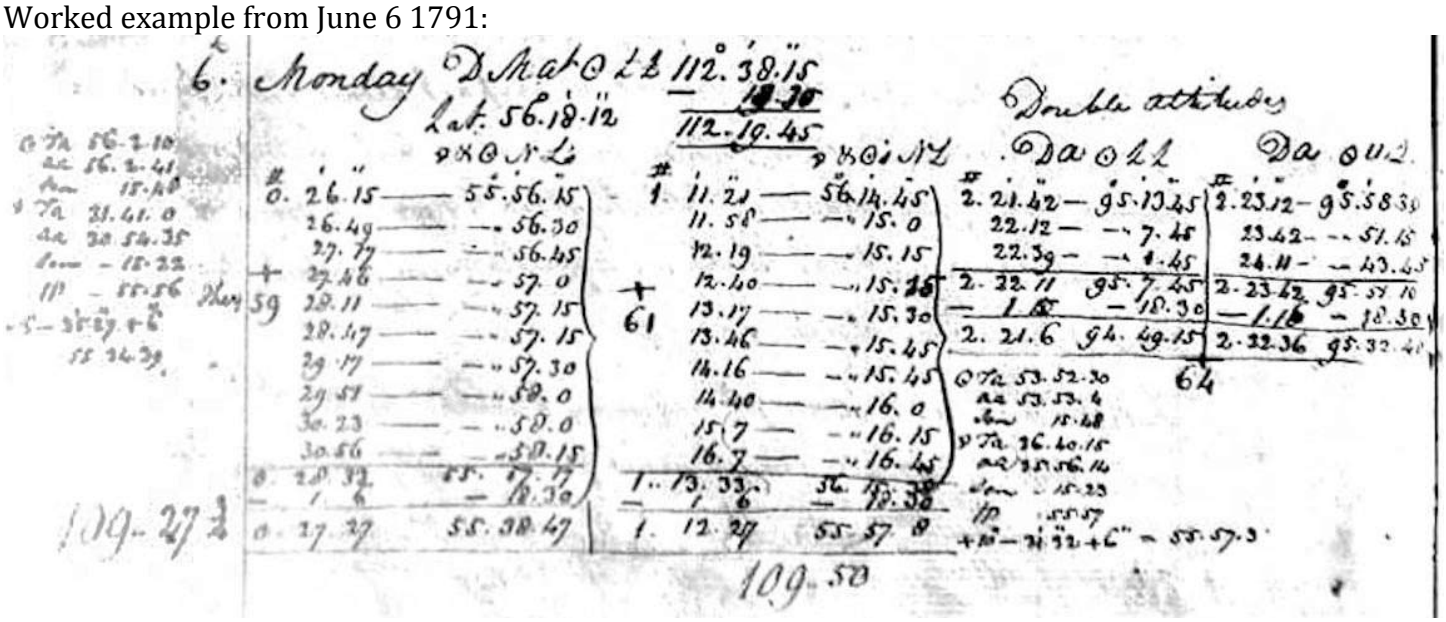
Basic Procedure for finding latitude and longitude

- start from a known lat,lon, set watch to local apparent time
- take courses during travel (magnetic)
- convert courses to true
- update lon by account (from courses)
- measure index error of sextant with sun
- measure sun meridian altitude at noon using artificial horizon (AH – pool of mercury in a dish) with parallel glasses (a glass frame over the dish to prevent wind ripples; the glass must have parallel faces (this was hard to make before float glass was invented))
    - Before starting, the approximate latitude and sun declination is used to pre-set the sextant to double the expected altitude (or he just looks in his notebook to see the meridian angle from a day or two ago)
- measure distance between sun (near limb) and moon (sharp limb)
    - The approximate lunar distance in the nautical almanac is used to pre-set the sextant so that the two bodies are easily found by sweeping the sky
    - sextant measures angle between the limbs of the bodies, pocket watch is used to record the time of each measurement
    - Fidler uses 10 sights, as close together as can be done (about 30-45 sec between sights)
- about 2 or 3 hours later (or earlier if more convenient) take time sight
    - the sextant is pre-set to double the expected altitude
    - sextant measures altitude of sun by touching the lower (LL) or upper limbs (UL) of the sun with it's reflection in an AH
    - Fidler, Turnor, Thompson all use 3 sights of LL and 3 of UL and compute the time correction for each run separately
- do calculations on these data
    - latitude
        - subtract index error from double alt and divide by 2
        - subtract refraction based on altitude (and temperature if far from 10°C)
        - use long. by account to get approximate Greenwich apparent time (GAT)
        - look up sun's declination (sun_dec) and semidiameter (sun_sd) at GAT in the nautical almanac for this date
        - lat = polar distance (PD, 90°) – (alt +/- sun_sd) +/- sun_dec
            - if lower limb of sun is used, then add sun_sd
            - if sign of sun_dec (N or S) is same as lat, then add sun_dec, else subtract
    - time sight
        - average times and double altitudes for each run of LL and UL
        - remove index error from altitude, divide by 2, add sun_sd, subtract refraction to get H_o
        - Spherical trig gives the following relation for local hour angle (LHA – angular separation of the sun from the meridian)
            - Cos(LHA) = [(sin(H_o) – sin(sun_dec) * sin(lat))/(cos(sun_dec) * cos(lat))]
        - Sun_dec was obtained for the latitude at noon, now it has to be advanced to the time of the time sight by proportioning based on the rate of change in the almanac
        - Follow algorithm for solving this equation using log tables
        - Convert LHA to time and the difference between the avg. time of time sight and correct time is the watch error.
    - Lunar distance

- The angular separation of the sun and the moon must be corrected for the effects of refraction on both bodies and the parallax of the moon (the moon is closer to the observer when directly overhead then when on the horizon). This is done by finding the components of each effect acting on the line between the bodies and making corrections. In the 18th century, it was common to use a series solution with two linear terms and one quadratic term, adapted for the use of log tables. Fidler, Turnor, Thompson always use Witchell's method for clearing the distance.
- The cleared distance is then compared with the bracketing distances in the nautical almanac and proportioned to find the exact value of GAT that corresponds to the cleared distance.
- The time correction from the time sight is applied to the average time of the lunar distance observation to find the exact local time of the lunar sight.
- The difference between GAT and LAT converted to degrees (15 degrees per hour) is the longitude

Fidler tries to take a meridian altitude of the sun every day at noon if the sky is clear. He only takes lunar distances and time sights after traveling for several days so that he can reset his survey by courses. The following worked example of sights for latitude and longitude show the process he used to convert his sextant observations into a location on the earth. If you are not familiar with working with log tables, then these methods will seem odd. There are many online tutorials that will orient those who went to school after the 1970s (basic instructions for log tables were dropped from the high school curriculum in the late 70s and they had stopped being used everywhere long before that time). To learn the basic principles behind the lunar distance method for finding longitude, an excellent tutorial is at https://reednavigation.com/lunars/easylunars/. I won't give any theory here, just a rote re-working of a particular lunar shot by Peter Fidler in 1791.

Worked example from June 6 1791:



This is Fidler's compact summary of a double meridian altitude for latitude sight at noon, two lunar distance sights between the sun and the moon just after noon, two time sights at about 2:20 and 2:30 in the afternoon, and the associated almanac data and calculated values used to work up the observations to get latitude and longitude.

He begins with a noon sight to determine the meridian altitude of the sun (the maximum angle that is formed between the sun and the horizon). Since he is not on the ocean, he uses a pool of mercury to reflect the sun (gravity makes the surface level, thus parallel with a line extending to a hypothetical sea

horizon) and measures the angle between the real sun and the reflected sun, bringing the lower edge of the real sun to just touch the upper edge of the reflected sun in the superimposed field of the sextant. This is what is meant by his notation of D M alt ⊙ LL 112°.38'.15". He would probably have first found the index error of his sextant by touching the real sun to itself in both the direct and reflected views of his sextant to find out what reading corresponded to what should be a zero reading (he actually would touch the sun's limbs to each other and correct for the semidiameter as superimposing the sun on itself is not very accurate). The index error is then removed from the double altitude measurement (in this case it is a subtraction of 18'30" of arc). He then needs the sun's declination and semidiameter for the time of measurement from the nautical almanac, but to do this he needs to know GAT at the time of local noon. It would be a circular problem but for the fact that he is not lost and has kept track of his East-West travel distances and knows his approximate longitude (we will assume that his longitude by account is 109°W thus GAT is 109/15 = 7h:16' later than local time. He looks up the sun's position in the almanac and finds:

[62] JUNE 1791. II.

Days of the Month.	Days of the Week.	Sun's Longitude.	Sun's Right Asc. in Time.	Sun's Declin. North.	Equat. of Time. Sub.	Diff.
		S. D. M. S.	H. M. S.	D.M. S.	M. S.	S.
1	W.	2. 10.49.58	4. 36.59,3	22. 5.31	2.37,8	9,2
2	Th.	2. 11.47.26	4. 41. 5,1	22.13.26	2.28,6	9,6
3	F.	2. 12.44.53	4. 45.11,3	22.20.58	2.19,0	9,9
4	Sa.	2. 13.42.18	4. 49.17,8	22.28. 6	2. 9,1	10,2
5	*Su.*	2. 14.39.42	4. 53.24,6	22.34.51	1.58,9	10,6
6	M.	2. 15.37. 5	4. 57.31,8	22.41.12	1.48,3	11,0
7	Tu.	2. 16.34.28	5. 1.39,3	22.47. 9	1.37,3	11,2
8	W.	2. 17.31.49	5. 5.47,1	22.52.42	1.26,1	11,4
9	Th.	2. 18.29. 9	5. 9.55,1	22.57.51	1.14,7	

These values are for noon on the appointed days, the start of the astronomical day that was used in the Nautical Almanac (NA). He finds that the sun's declination is 22°41'12" on June 6 and 22°47'9" on June 7, so he has to proportion between these values as 7h16' proportions between 0 and 24 hours. Logarithms were used to avoid multiplication and division and Maskelyne had developed a table of proportional logs specifically to find the proportional value within 3 hours as the lunar distance tables are divided into 3 hour blocks. For proportioning the daily values that change more slowly, the same table can be used by looking at hours as minutes. So to find the value of declination corresponding to 7h16'/24h by 5'57" difference we add the proportional log (plog) of 7'16" to plog of 5'57" and subtract the plog of 24' to get 1.3939 + 1.4808 – 0.8751 = 1.9996. Reversing this value we get 1'49" and add that to the starting declination of 22°41'12" to get a declination of 22°43'1". In order to calculate latitude, we also need the sun's semidiameter for the date and that comes from a different table in the NA:

III. JUNE 1791. [63]

Days.	Semidiameter of the Sun.	Time of D° passing the Meridian.	Hourly Motion of the Sun.	Logarithm of the Sun's Distance.	Place of the Moon's Node.
	M. S.	M. S.	M. S.		S. D. M.
1	15. 48,8	1. 8,3	2. 23,6	0. 006346	6. 19. 17
7	15. 48,2	1. 8,6	2. 23,4	0. 006657	6. 18. 58
13	15. 47,6	1. 8,7	2. 23,2	0. 006890	6. 18. 39
19	15. 47,2	1. 8,8	2. 23,1	0. 007074	6. 18. 20
25	15. 47,0	1. 8,8	2. 23,0	0. 007204	6. 18. 1

We can take the value from June 7 of 15'48" since 10ths of a second won't matter here. The latitude is then found by taking 90 (the zenith distance) and subtracting the sun's altitude and adding the declination (since it is North and the latitude is also North) to get 56°18'0". This is close to Fidler's value of 56°18'12" (the true latitude of the place is 56°19'36"); the difference is probably just the proportioning of declination.

Having measured his latitude, and seeing that the moon is visible in the sky and within a reasonable distance to take a measurement (between 30°-120° of separation), he prepares to measure the distance between the sun and moon's near limbs (the moon is a waxing crescent East of the sun, so he will just touch the near limbs to measure the distance). He has pre-set his sextant to about 56° so it is a simple matter to get the two bodies in sight and bring them to just touch, when he will note the local time on his pocket watch and write down the time and distance. He does this 10 times and then waits a couple of hours until he can take a time sight (the change in altitude with time is greatest as a body is rising in the East or setting in the West, so that is when the time can be most accurately measured). In truth, he can't spend all day sitting around, so after a couple of hours the sun has moved far enough West to take a time sight. He measures the double altitude through his AH first using the sun's lower limb and then using the sun's upper limb. He needs to find the correction for his watch before he can work his lunar so that he will know the exact local apparent time of his lunar distance measurement.

The time sight is analyzed just like the latitude measurement except he has taken 3 separate measurements of each limb and noted the time of each. He then averages the times and double altitudes and proceeds to analyze the average values. As with latitude he corrects for index error, divides the double altitude by 2 and then adjusts for refraction and semidiameter to get the proper altitude. He notes that the sun's declination is increasing at about 15"/h and so advances his noon value to 22°43'36" for 2 1/3 hours. From spherical trigonometry, the time is related to the sun's altitude by:
Cos(LHA) = [(sin(H_o) – sin(dec) * sin(lat))/(cos(dec) * cos(lat))]
Where LHA is local hour angle, H_o is the true altitude of the sun, and the others we have already met. Since this is a hard calculation to do by hand, Fidler will follow an algorithm designed for log tables.

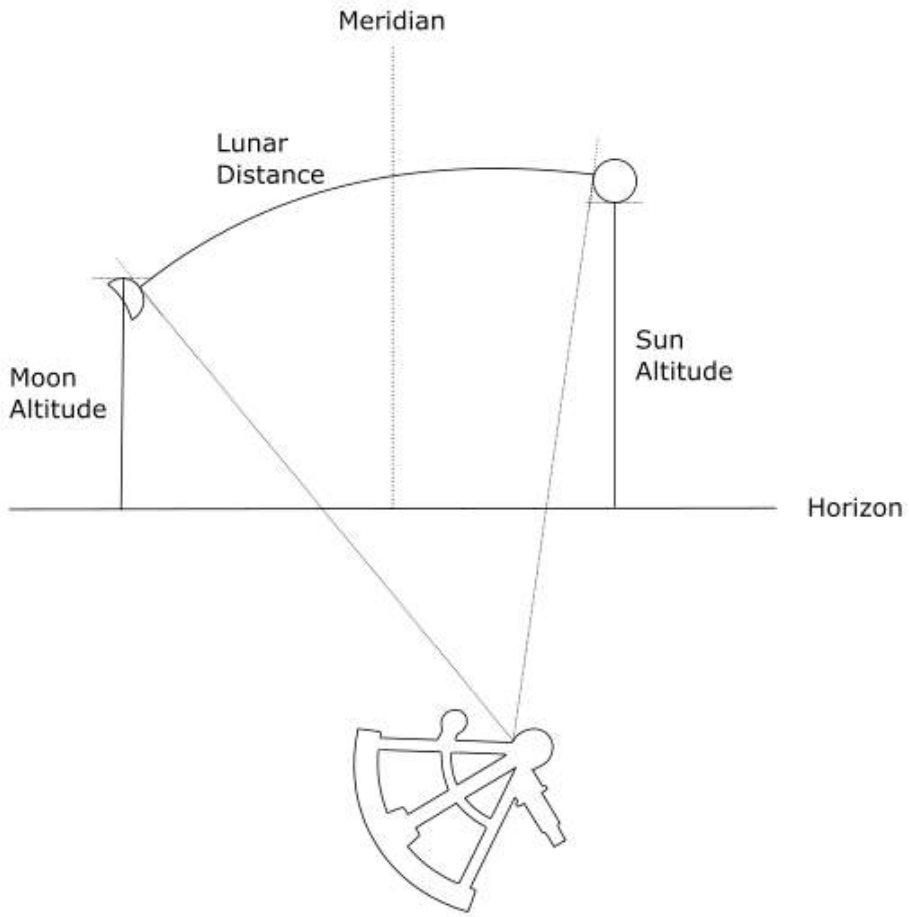
Summary of algorithm:
A=(latde + H_o)/2
a=log cot(A)
b=log tan(H_o~A) [subtract lesser from greater]
C=(90-dec)/2
c=log cot(C)
D=atan(a+b+c-20)
e=log tan(C+/-D) [if latde > H_o then subtract, else add]

LAT=acos(latde+e-10)*4

Fidler's first time sight on June 6 worked:
latde = 56°18'12"
sun dec = 22°43'41"
H_o = 47°40'2" [dbl alt – index error / 2 – refraction + sd]
Time = 2h22'11" [as per watch]
A = 51°59'57"
a = 9.89304
b = 8.87798
C = 33°38'39"
c = 10.17698
D = 5°4'10"
e = 9.73596
LAT = 2h21'3" (Fidler gets 2h21'6")

The time correction is then found by determining whatever has to be added to or subtracted from the watch reading to get LAT. For Fidler this is a subtraction of 1'5". For his other time sight he gets a correction of -1'6" and this is the correction he uses for his lunar distance calculation.

Lunar Distance Measurement



The lunar distance is the angle between the moon and the sun (or a star or planet). It is complicated because the true position of the bodies is altered by refraction and (for the moon) parallax as it approaches the horizon. By finding the effects of these two factors on the altitudes of the two bodies, the apparent distance can be cleared of those effects to give the true distance. The clearing process finds the component change along the arc from the moon to the other body and adds or subtracts these corrections

to give the true lunar distance. The Nautical Almanac then gives the Greenwich time that corresponds to that exact distance.

Fidler has not measured the altitudes of either the sun or the moon at the time of taking his lunar distance, but he needs to know them to account for refraction and parallax. He can reasonably easily get the true altitudes from the NA since he knows local time (which is just the hour angle of the sun) and can get the right ascensions (RA) of both the sun and moon from the NA. Then he works backwards to add in the effects of refraction and parallax to get the apparent altitudes for both bodies. Fidler, Turnor, and Thompson always calculate the altitudes in this way; there is no instance where they measure an altitude for working a lunar.

The first step is to get RA and Dec for the sun and the moon by proportioning the almanac values to the time of the lunar (local time + longitude by account converted to time). The local time of the lunar is 12h27m27s PM and we are using 109° as longitude by account (= 7h16m) so GAT is 7h43m27s on June 6.

[66] JUNE 1791. VI.

Days of the Month.	Days of the Week.	☽'s Age.	☽'s Passage over Merid.	☽'s Right Ascens. at Noon.	☽'s Right Ascens. at Midn.	☽'s Declination at Noon.	☽'s Declination at Midn.
			H. M.	D. M.	D. M.	D. M.	D. M.
1	W.	1	6	65. 26	71. 39	17. 49 N	18. 20 N
2	Th.	2	0. 32	77. 55	84. 13	18. 39	18. 46
3	F.	3	1. 20	90. 33	96. 53	18. 40	18. 21
4	Sa.	4	2. 8	103. 13	109. 33	17. 50	17. 6
5	Su.	5	2. 56	115. 52	122. 10	16. 10	15. 3
6	M.	6	3. 44	128. 27	134. 42	13. 44	12. 16
7	Tu.	7	4. 31	140. 57	147. 11	10. 38	8. 52
8	W.	8	5. 18	153. 26	159. 43	6. 58	4. 58
9	Th.	9	6. 6	166. 3	172. 26	2. 53 N	0. 44 N

The moon moves across the sky faster than the sun so the NA gives values for every 12 hours rather than 24 for the sun. The moon's RA at noon on the 6th is 128°27' and at midnight is 134°42'. The difference is 6°15' over 12 hours so we proportion by logs by subtracting the log of 7h43m27s (10.88785) from the log of 12h (11.07918) and then subtracting that (0.19133) from the log of 6°15' (0.79588) to get 0.60455 which backs out of the log tables to 4°1'22" which is added to the June 6th baseline to get the moon's RA at the time of the lunar as 132°28'22". A similar operation is used to get values for the sun's RA, and Dec:

	Sun	Moon
RA	74°42'51"	132°28'22"
Dec	22°43'06"	12°47'21"

We also need the semidiameters of the sun and moon and the horizontal parallax of the moon. For the moon's values, another page in the NA is consulted:

VII. JUNE 1791. [67]

Days of the Month.	Days of the Week.	Semidr. ☽ at Noon. M. S.	Semidr. ☽ at Midnight. M. S.	Hor. Par. ☽ at Noon. M. S.	Hor. Par. ☽ at Midnight. M. S.	Proport. Logar. at Noon.	Proport. Logar. at Midn.
1	W.	14. 45	14. 46	54. 7	54. 10	5219	5215
2	Th.	14. 47	14. 49	54. 15	54. 22	5209	5199
3	F.	14. 51	14. 54	54. 30	54. 39	5189	5177
4	Sa.	14. 57	15. 0	54. 50	55. 1	5162	5148
5	Su.	15. 3	15. 7	55. 14	55. 29	5130	5111
6	M.	15. 11	15. 16	55. 45	56. 2	5090	5068
7	Tu.	15. 21	15. 27	56. 21	56. 41	5044	5018
8	W.	15. 33	15. 39	57. 2	57. 25	4991	4962
9	Th.	15. 45	15. 52	57. 49	58. 13	4932	4902

The moon's semidiameter is augmented by an amount due to parallax which is 8" in this case.

	Sun	Moon
SD	15'48"	15'22"
HP		55'56"

The calculation of the sun's true altitude is straight forward using local apparent time converted to an hour angle (LAT)*15 = 6°51'47". The log secant of this angle (10.0031) is added to the log tangent of the sun's dec (9.58682) and after subtracting 10 is taken back to an angle A = 22°51'53". Declination and Latitude are both North in this case so we subtract A from the latitude to get an angle B = 33°26'17". The true altitude is then the log sin of B added to the log cosecant of A and the log sin of Dec, subtracting 20 and backing out of the log table to get TA = 56°02'21". The table for refraction give 40" for this angle so we add that to get and apparent altitude AA = 56°03'01". We can see Fidler's values in the left margin as TA = 56°02'10" and AA = 56°02'41"; the small differences are probably due to his longitude of account being slightly different from our guess.

The moon's altitudes are calculated in the same way except the hour angle is given by 360 minus the sun's hour angle plus the sun's RA plus the moon's RA. This gives the moon's HA as 309°06'17" and thus TA = 31°42'01". The moon's apparent altitude is affected by both refraction and parallax. For refraction we add 1'36" and subtract the parallax in altitude (cos(alt) * HP) = 47'35" to get AA = 30°56'02". These values are again almost identical to Fidler's.

Now the lunar distance can be cleared for the effects of parallax and refraction to get the true lunar distance from the observed distance (55°38'46") corrected for center to center by adding the two semidiameters (56°09'56"). Witchell's method gives 3 corrections, found by following a recipe from a navigation manual (this one is from Moore's New Practical Navigator as it's easier to follow than Witchell's description).

> First add the sun or star's and moon's apparent altitudes together, and take half the sum; then subtract the less from the greater, and take half the difference; then add together:
>
> > The cotan of half the sum,
> > The tan of half the difference, and
> > The cotan of half the apparent distance,
>
> Their sum, rejecting 20 in the index, will be the log tan of an angle A.

30°56'02" + 56°03'01" = 86°59'04" / 2 = 43°29'32"

56°03’01” - 30°56’02” = 25°06’58” / 2 = 12°33’29”
log cotan (43°29’32”) = 10.02287
log tan (12°33’29”) = 9.34784
log cotan (28°04’58”) = 10.27281
Angle A = arctan (inverse log(9.64352)) = 23°45’09”

Second, when the sun or star’s altitude is greater than the moon’s, take the difference between A and half the apparent distance, but if less, take their sum, then add together:

The cotan of this sum or difference,
The cotan of the sun or star’s apparent altitude, and
The proportional log of the correction of the sun or star’s altitude;

Their sum, rejecting 20 in the index, will be the proportional log of the 1st correction.

28°04’58” - 23°45’09” = 4°19’49”
log cotan (4°19’49”) = 11.12078
log cotan (56°03’01”) = 9.828162
plog (40”) = 2.4314
inverse plog (3.3803) = 5” (first correction)

Third, if the sum of A and half the apparent distance was taken in the last article, take now their difference; but if their difference, take now their sum. Then add together:

The cotan of their sum or difference,
The cotan of the moon’s apparent altitude, and
The proportional log of the correction of the moon’s apparent altitude.

Their sum, rejecting 20 in the index, will be the proportional log of the 2nd correction.

28°04’58” + 23°59’06” = 52°04’04”
log cotan (52°04’04”) = 9.89175
log cotan (30°56’02”) = 10.22236
plog (45’58”) = 0.5928
inverse plog (0.7069) = 35’21” (second correction)

Fourth, when A is less than half the apparent distance, the 1st correction must be added to, and the 2nd correction subtracted from the apparent distance; but when A is greater, their sum must be added to the apparent distance, when the sun or star’s altitude is less than the moon’s; but when the moon’s altitude is less, their sum must be subtracted to give the corrected distance.

The first correction is added, the second correction is subtracted from the apparent distance
56°09’56” + 05” – 35’21” = 55°34’40” (distance with the first two corrections applied)

Fifth, in table X, look for this last corrected distance in the top column, and the correction of the moon’s altitude in the left-hand side column; take out the number of seconds that stand under the former and opposite to the latter. Look again in the same table for the corrected distance in the top column, and the principal effect of the moon’s parallax in the left hand side column, and take out the number of seconds that stand under the former and opposite the latter. The difference between these 2 numbers must be added to the corrected distance if less than 90°, but subtracted from it if more than 90°; the sum or difference will be the true distance.

Correction due to the moon’s altitude is 35’21”, the effect of parallax is 47’45”
The table gives values of [((*difference*^2)/2) * cotan(*corrected distance*)] / 60 where *difference* is in minutes of arc, *corrected distance* is in degrees, and the result is in seconds of arc.
Table X (*difference* =35’, *corrected distance* =56°) = 7”
Table X (*difference* =48’, *corrected distance* =56°) = 13”
The difference is 6” and it is to be added to the corrected distance (3rd correction)
The corrected distance is thus 55°34’46” and the corrections, which Fidler writes in the margin, are:
+5” -35’21” + 6” (Fidler’s corrections are +5” – 35’27” +6” and he writes 55°34’39” as his cleared distance).

The corrected distance is then looked for in the NA to find Greenwich Apparent Time

Diftances of ☽'s Center from Sun, and from Stars weft of her.

Days.	Stars Names.	Noon.	3 Hours.	6 Hours.	9 Hours.	12 Hours.	15 Hours.	18 Hours.	21 Hours.
		D. M. S.	D. M. S.	D. M. S.	D.M.S.	D. M. S.	D. M. S.	D. M. S.	D. M. S.
5	The Sun.	40. 29. 5	41. 53. 41	43. 18. 28	44.43.26	46. 8. 35	47. 33. 56	48. 59. 30	50.25.17
6		51. 51. 16	53. 17. 29	54. 43. 56	56.10.37	57. 37. 32	59. 4. 41	60. 32. 5	61.59.44
7		63. 27. 37	64. 55. 47	66. 24. 13	67.52.55	69. 21. 54	70. 51. 10	72. 20. 43	73.50.35
8		75. 20. 43	76. 51. 10	78. 21. 55	79.53. 0	81. 24. 23	82. 56. 7	84. 28. 11	86. 0.36
9		87. 33. 20	89. 6. 25	90. 39. 50	92.13.37	93. 47. 44	95. 22. 14	96. 57. 7	98.32.20
10		100. 7. 56	101. 43. 54	103. 20. 13	104.56.55	106. 33. 58	108. 11. 24	109. 49. 13	111.27.23
11		113. 5. 55	114. 44. 49	116. 24. 4	118. 3.40	119. 43. 38			
9	Regulus.	19. 23. 5	21. 1. 0	22. 39. 33	24.18.42	25. 58. 30	27. 38. 52	29. 19. 45	31. 1. 9
10		32. 43. 6	34. 25. 28	36. 8. 17	37.51.33	39. 35. 17	41. 19. 26	43. 4. 1	44.49. 1
11		46. 34. 26	48. 20. 16	50. 6. 29	51.53. 6	53. 40. 7	55. 27. 31	57. 15. 17	59. 3.25
12		60. 51. 55	62. 40. 46	64. 29. 57	66.19.27	68. 9. 18	69. 59. 28	71. 49. 55	73.40.41
13		75. 31. 44							
13	Spica ♍	21. 53. 35	23. 44. 46	25. 36. 11	27.27.54	29. 19. 56	31. 12. 12	33. 4. 39	34.57.17
14		36. 50. 8	38. 43. 7	40. 36. 12	42.29.24	44. 22. 43	46. 16. 8	48. 9. 35	50. 3. 3
15		51. 56. 34	53. 50. 4	55. 43. 30	57.36.54	59. 30. 15	61. 23. 31	63. 16. 39	65. 9.39
16		67. 2. 32	68. 55. 16	70. 47. 46	72.40. 5	74. 32. 12			
16	Antares.					29. 51. 51	31. 39. 25	33. 27. 2	35.14.43
17		37. 2. 30	38. 50. 17	40. 37. 56	42.25.29	44. 12. 58	46. 0. 13	47. 47. 12	49.33.56
18		51. 20. 28	53. 6. 40	54. 52. 31	56.38. 3	58. 23. 17	60. 8. 10	61. 52. 39	63.36.45

We can see the bracketing values of D1: 54°43’56” at 6 Hours on the 6th and D2: 56°10’37” at 9 Hours. To proportion the time after 6PM we use proportional logs:

Cleared distance – D1 = 55°34’46” - 54°43’56” = 50’50”

D2 – D1 = 56°10’37”- 54°43’56” = 1°26’40”

Proplog(50’50”) = 0.5491

Proplog(1°26’40”) = 0.3174

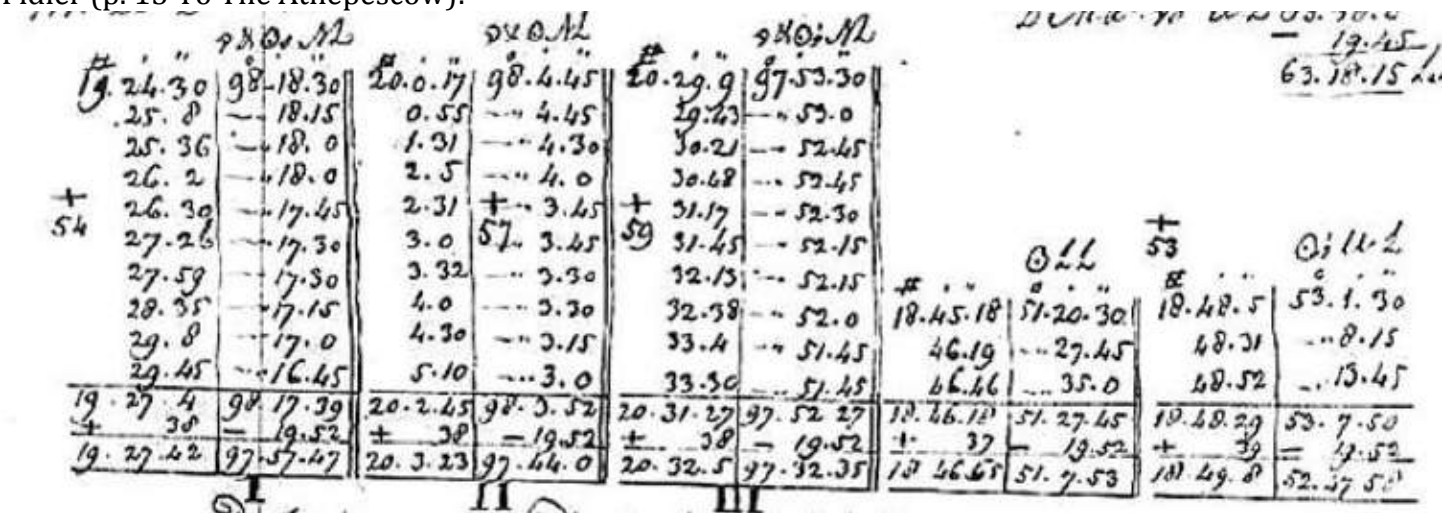
Inverse proplog of difference (0.2317) = 1h45’34”

We add 1h45’34” to 6 hours to get 7h45’34” GAT, subtracting LAT of 0h27’27” gives a time difference of 7h18’07”. Converting to longitude by multiplying by 15 gives 109°31’45”. Fidler writes 109..27 ¼ in the margin; close enough. Fidler shot another lunar just after 1:00 PM and worked it up in the same way to get a longitude of 109..50. The actual longitude is 109°56’45” so his average is out by a little over 6 miles: a very good result. The theory of the moon’s motion wasn’t all that good until about 1830 and these old almanacs can have pretty large errors in the lunar distances. Using a simulation of the solar system done by the JPL, we can now calculate lunars more accurately for these past eras. In this particular case, the lunar distance for the sun on June 6, 1791 is 16” too large, leading to an error in time of about 30” and therefore of about 7 ½ minutes of longitude (about 5 miles in this case). With a better almanac, Fidler would have been almost dead on (but just by luck, a handheld sextant can’t actually measure that accurately in a repeatable fasion).

Fidler vs Turnor navigation methods

In the introductory essay to “Peter Fidler: From York Factory to the Rocky Mountains”, Barbara Belyea suggests that Peter Fidler knew celestial navigation before he started working for the HBC; that he wasn’t trained in the art by Turnor as David Thompson was. There is a particular trick that Fidler uses when taking lunar distances that seems to support that hypothesis since Philip Turnor does not use this trick before he travels with Fidler (but he does use it when they are travelling together and afterward as well) and David Thompson never uses it. It’s just a method for simplifying the averaging of times and angles; not a huge improvement to the technique of lunar distances, but anything that makes sexagesimal arithmetic a bit easier is worth knowing. Fidler’s technique is to always take 10 measurements of the lunar distance so that he can use a right shift of the sum to perform a division by 10 (needed to average the sample). It is not a method that would ever be used on a ship since the measurements themselves can be tricky on a moving platform, but lends itself nicely to land-based surveying. Here is an example from Fidler’s journal of 1791:

June 21 1791
Fidler (p. 15 To The Athepescow):



To average the left hand time column:

19	24	30
19	25	8
19	25	36
19	26	2
19	26	30
19	27	26
19	27	59
19	28	35
19	29	8
19	29	45
190	266	279
19	27	28
	borrow 4	4

The left column is straightforward, but using the rule, we add to 190 and shift right 1 to get 19. The second column sums to 266, we add 4 to make a number divisible by 10 and shift right 1 to get 27. The 3rd column sums to 279, we subtract 4 * 60 (4 was borrowed) to get 39 which we round to 40 and shift right

1 to get 4. The average is 19h27’04”. At the same time, Turnor takes his own lunar distances and also takes 10 sights for each of two runs:
Turnor (p 391 Champlain Society, JB Tyrell ed): _____

Ther	Time			Doub. Alt ⊙ L L			Ther	Time			⊙ L L Doub. Alt		
+ 50	18	3	32	39	54		+ 57	20	16	28	75	54	
		4	12	40	5				17		76	3	
		4	37	40	11	15			17	30		11	
	18	4	7	40	3	25		20	16	59	76	2	40
Error		+	39	−	19	52			+	38	−	19	52
	18	4	46	39	43	33		20	17	37	75	42	48

Ther-mom-iter	Time by Watch and reduced to apparent			Distances			Longi-tude	Ther-mom-iter	Time by Watch and reduced to apparent			Distances			Longi-tude
+	H	′	″	⊙&☽ N L				+	H	′	″	⊙&☽ N L			
55	19	34	4	98	15	30		57	20	23	43	97	55	15	
		34	31		15	15				24	2		55		
		34	52		15					24	30		55		
		35	18		15					24	51		54	45	
		35	42		14	45				25	18		54	30	
		36	8		14	30				25	45		54	30	
		36	32		14	15				26	5		54		
		37	10		14					26	29		53	45	
		37	37		13	45				26	50		53	45	
		38	14		13	30				27	10		53	30	
	19	36	1	98	14	33			20	25	28	97	54	24	
		+	38	−	19	52				+	38	−	19	52	
	19	36	39	97	54	41	111°13′		20	26	6	97	34	32	111°28¼′

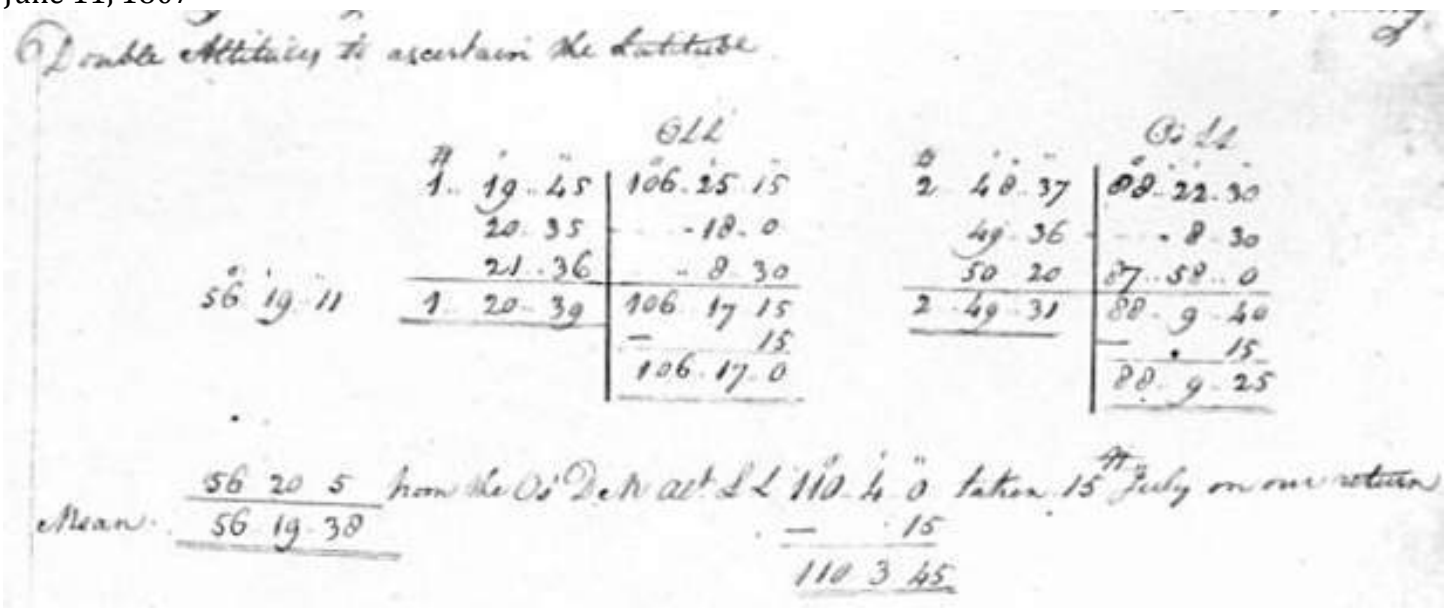
They seem to be sharing the same watch since their measurements are staggered. Odd that they don’t include each other’s results (unless they were no longer together when they worked the lunars).

In “Result of astronomical observations, made in the interior parts of North America”, Turnor most frequently uses 5 or 6 lunar distances prior to 1791.

David Thompson almost always uses 5-8 measurements; never 10. It should be noted that in Fidler’s astronomical journals kept during his stay at various posts, he often uses 8 sights rather than 10, but when he is travelling, 10 sights are almost always taken.

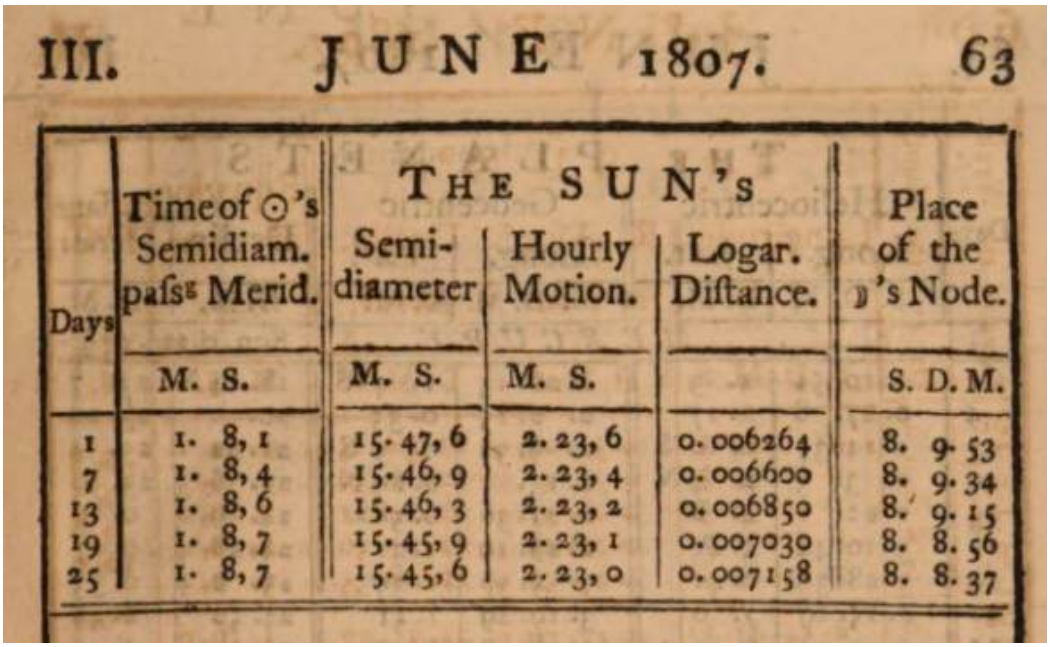
Fidler also uses other techniques for finding latitude and longitude on occasion. Some examples of these methods are shown below:

Latitude by Double Altitudes
June 11, 1807


Double Altitude to ascertain the Latitude

	☉ ⊥		☉ ⊥		
	1st 19 45	106 25 15		2 40 37	88 22 30
	20 35	-18 0		49 36	-8 30
	21 36	-8 30		50 20	87 58 0
56 19 11	1 20 39	106 17 15		2 49 31	88 9 40
		— 15			— 15
		106 17 0			88 9 25

56 20 5 from the ☉'s Decl. at ⊥ 110 4 0 taken 15th July on our return
Mean 56 19 38 — 15
110 3 45


III. JUNE 1807. 63

Days	Time of ☉'s Semidiam. passg Merid.	THE SUN'S Semi-diameter	Hourly Motion.	Logar. Distance.	Place of the ☽'s Node.
	M. S.	M. S.	M. S.		S. D. M.
1	1. 8, 1	15. 47, 6	2. 23, 6	0. 006264	8. 9. 53
7	1. 8, 4	15. 46, 9	2. 23, 4	0. 006600	8. 9. 34
13	1. 8, 6	15. 46, 3	2. 23, 2	0. 006850	8. 9. 15
19	1. 8, 7	15. 45, 9	2. 23, 1	0. 007030	8. 8. 56
25	1. 8, 7	15. 45, 6	2. 23, 0	0. 007158	8. 8. 37

The sun's semidiameter is 15'46".

Days of the Week.	Days of the Month.	THE SUN'S Longitude.	R'. Ascen. in Time.	Declin. North.	Equation of Time. Sub.	Diff.
		S. D. M. S.	H. M. S.	D. M. S.	M. S.	S.
M.	1	2. 9. 59. 59	4. 33. 25, 9	21. 58. 15	2. 44, 1	8, 8
Tu.	2	2. 10. 57. 27	4. 37. 31, 3	22. 6. 30	2. 35, 3	9, 3
W.	3	2. 11. 54. 54	4. 41. 37, 1	22. 14. 21	2. 26, 0	9, 6
Th.	4	2. 12. 52. 21	4. 45. 43, 3	22. 21. 50	2. 16, 4	10, 0
F.	5	2. 13. 49. 46	4. 49. 50, 0	22. 28. 55	2. 6, 4	10, 4
Sa.	6	2. 14. 47. 11	4. 53. 56, 9	22. 35. 37	1. 56, 0	10, 7
Sun.	7	2. 15. 44. 35	4. 58. 4, 2	22. 41. 55	1. 45, 3	11, 0
M.	8	2. 16. 41. 58	5. 2. 11, 8	22. 47. 49	1. 34, 3	11, 3
Tu.	9	2. 17. 39. 20	5. 6. 19, 6	22. 53. 19	1. 23, 0	11, 5
W.	10	2. 18. 36. 41	5. 10. 27, 7	22. 58. 25	1. 11, 5	11, 7
Th.	11	2. 19. 34. 2	5. 14. 36, 1	23. 3. 6	0. 59, 8	12, 0
F.	12	2. 20. 31. 21	5. 18. 44, 6	23. 7. 24	0. 47, 8	12, 1
Sa.	13	2. 21. 28. 39	5. 22. 53, 4	23. 11. 16	0. 35, 7	12, 3
Sun.	14	2. 22. 25. 57	5. 27. 2, 2	23. 14. 45	0. 23, 4	12, 5
M.	15	2. 23. 23. 13	5. 31. 11, 3	23. 17. 49	0. 10, 9	12, 5
Tu.	16	2. 24. 20. 29	5. 35. 20, 4	23. 20. 28	*Add.* 1, 6	12, 7
W.	17	2. 25. 17. 44	5. 39. 29, 7	23. 22. 42	0. 14, 3	12, 7
Th.	18	2. 26. 14. 58	5. 43. 39, 0	23. 24. 32	0. 27, 0	12, 9
F.	19	2. 27. 12. 12	5. 47. 48, 4	23. 25. 57	0. 39, 9	12, 8
Sa.	20	2. 28. 9. 25	5. 51. 57, 8	23. 26. 58	0. 52, 7	

The declination is found by proportioning between June 11th and June 12th based on the time of the sight and the longitude by account. We will use 103°W as his longitude so we need to get the declination for 7 hours after noon on the 6th. As earlier, we subtract the log of the time since noon (10.84510) from the log of 24 (11.38021) to get 0.53511 which is subtracted from the log of the change in declination (8.85531) which is then taken back out of the log table and added to the declination on the 11th to get 23°4'21" as the declination at the time of observation.

His latitude by account will be given by his last latitude measurement on the 9th (56°8') advanced by the approximately 14 miles of Northing in his courses since then to 56°20'.

His two observations are divided by 2 and then the refraction is subtracted and the sun's semidiameter added to get true altitudes of 53°23'31" at 1h20m39s and 44°19'57" at 2h49m31s. The elapsed time between measurements is 1h28m51s and the half elapsed time is 0h44m25s.

Recipe from the Tables Requisite:

> To the log secant of the latitude by account, add the log secant of the sun's declination: their sum, rejecting 20 from the index, is the log ratio.

Log sec lat by acct = 10.25621
Log sec dec = 10.03621
Log ratio = 0.29241

> From the natural sine of the greater altitude, taken out of table XVII, subtract the natural sine of the least altitude: find the log of the remainder, and write it under the log ratio.

Natural sin of greater alt = 80273
Natural sin of lesser alt = 69882
Log remainder = 4.01666

> With half the elapsed time enter Table XVI and, from the column of half elapsed time, take out the log answering to it, which is also to be set down under the log ratio.

Table XVI in the Tables Requisite has 3 columns labelled "log ½ elapsed time" given by LOG(1/SIN(RADIANS([1/2 elapsed time]*15))), "log middle time" given by DEGREES(ASIN(10^([log value]-5.3))) and "log rising" given by LOG((SIN(RADIANS(15*[time from noon]/2))^2))+5.3. The table goes from zero to 5 hours of time.

Log ½ elapsed time = 0.71520

> Add these three logs together, and look for their sum in Table XVI in the column of middle time; and, having found the log nearest to it, take out the time corresponding, put it under half the elapsed time, and subtract the lesser from the greater: their difference will be the time from noon when the greater altitude was taken.

Sum of 3 logs = 5.02427
Time backed out from log middle time = 2h8m1s
½ elapsed time = 0h44m25s
subtraction result = 1h23m25s

> With this time enter the table again, and, from the column of log rising, take out the log corresponding to it: from this log subtract the log ratio, and the remainder will be the log of a natural number, which being found in table XVIII and added to the natural sine of the greater altitude, will give the natural sine of the meridional altitude of the sun.

Log rising = 3.81703
Log ratio = 0.29241
Subtraction result = 3.52462
Antilog = 3347
Sin greater alt = 80273
Sum = 83620
Arsin = 56°44'26"
Polar distance = 90 - 56°44'26" = 33°15'33"
Latitude = PD + dec = 56°19' 54"

Fidler has written 56°19'11" in the margin; the slight difference is probably due to differences in lat and long by account and/or table values.

Longitude by Lunar Eclipse
May 9, 1808

	H. ′ ″	Long^e W^t
Beginning of Eclipse of Watch	11. 1. 0	99. 41. ½
Beginning of Total Darkness	11. 59. 30	99. 35½
End of Total Darkness	13. 33. 0	99. 29
End of the Eclipse	14. 32. 0	99. 14
Mean of the Whole		99. 29. 56

On May 9, 1808 Fidler uses the timing of a lunar eclipse to determine his longitude at Brandon House. Underneath he writes, "Very clear Calm night – Moon reddish but still visible – very good sight from Beginning to End – observed with an achromatic Telescope magfn 35 times".

SOLAR and LUNAR ECLIPSES
IN THE YEAR 1808.

May. 9. *MOON eclipsed, invisible at Greenwich.*

	H. M.
Beginning of the Eclipse	17. 55⅓
Beginning of total Darkness	18. 54
Ecliptic ☾	19. 39
Middle	19. 40⅔
End of total Darkness	20. 27⅓
End of the Eclipse	21. 26

Digits eclipsed, 18°, 48′, from South Side of ☉'s Shadow.

May 24, *SUN eclipsed, invisible at Greenwich.* D. H. M.
☌ at 23^h. 19′½, in Long. 2^s. 4°. 0′, ☽'s Lat. 1°, 9′¼ S.

Oct. 19. *SUN eclipsed, invisible at Greenwich.*
☌ at 4^h. 54′½, in Long. 6^s. 26°. 8′, ☽'s Lat. 1°. 28′⅚ S.

Nov. 2. *MOON eclipsed, invisible at Greenwich.*

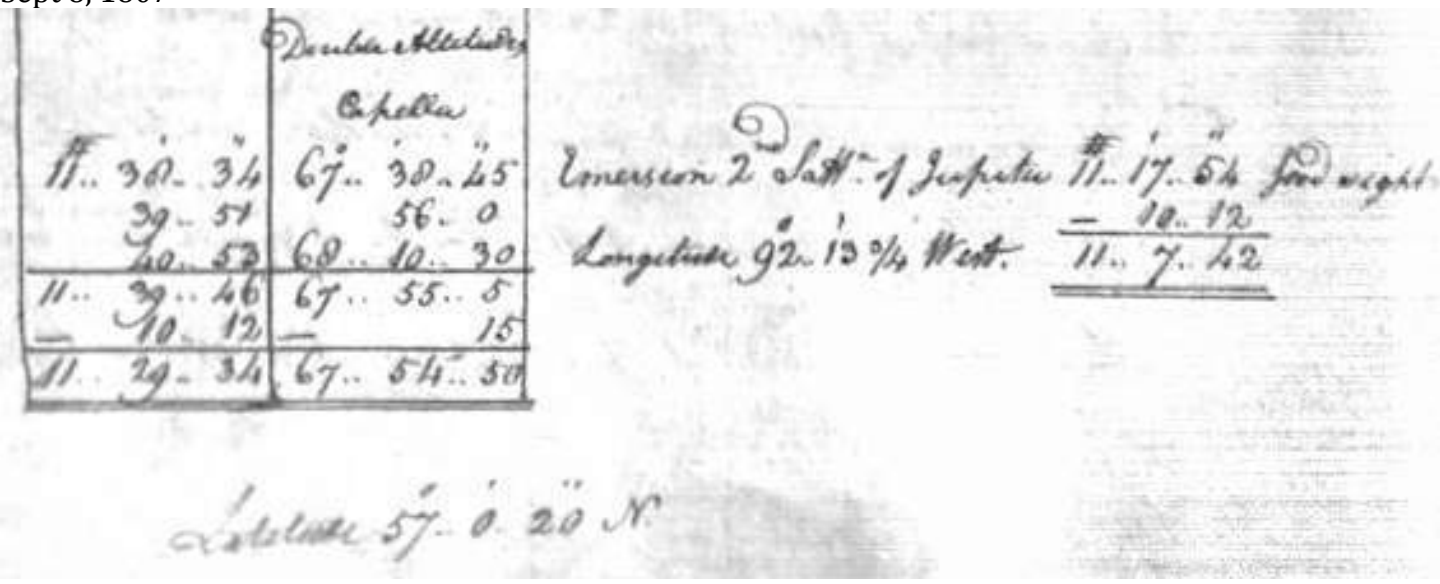
	H. M.
Beginning of the Eclipse	18. 32
Beginning of total Darkness	19. 39⅚
Ecliptic ☾	20. 26¾
Middle	20. 28⅔
End of total Darkness	21. 18
End of the Eclipse	22. 25⅓

Digits eclipsed, 18°. 9′, from North Side of ☉'s Shdw.

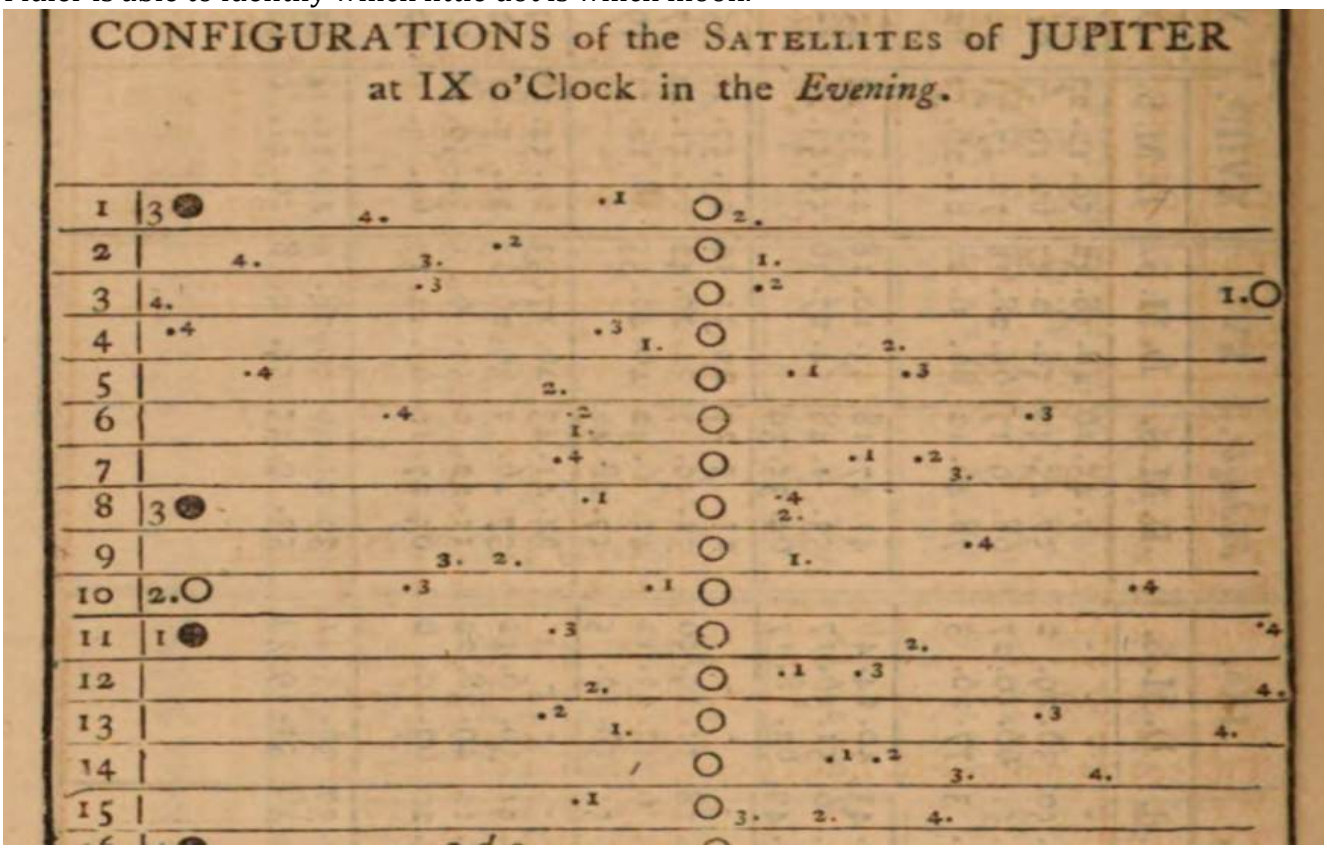
Nov. 17. *SUN eclipsed, invisible at Greenwich.*
☌ at 14^h. 55′¼, in Long. 7^s. 25°. 38′¼, ☽'s Lat. 1°. 13′. N.

Fidler simply subtracts his times from those in the almanac to get the longitude. For example, he finds the onset at 11h1m per his watch which is 6h54m36s different from the almanac value. Converted to longitude, this gives 103°39'. Elsewhere on the same page he records a watch error of 14m8s fast at 4:00 and 18m40s fast at 20:00 so a watch error of 15m50s at the time of the eclipse (11:00-14:00) is probably what he used to get 99°41 ¼' at that first time point. The actual longitude of Brandon House is 99°37' so he comes remarkably close. The rarity of lunar eclipses and the difficulty of observing them on a ship meant that they were rarely used for navigation.

Longitude by Jupiter's Moons
Sept 8, 1807



The almanac gives the orientation of Jupiter's 4 principal moons for each day of the month, from which Fidler is able to identify which little dot is which moon.



The almanac also gives the GAT for any eclipses of the moons on the days for which they occur:

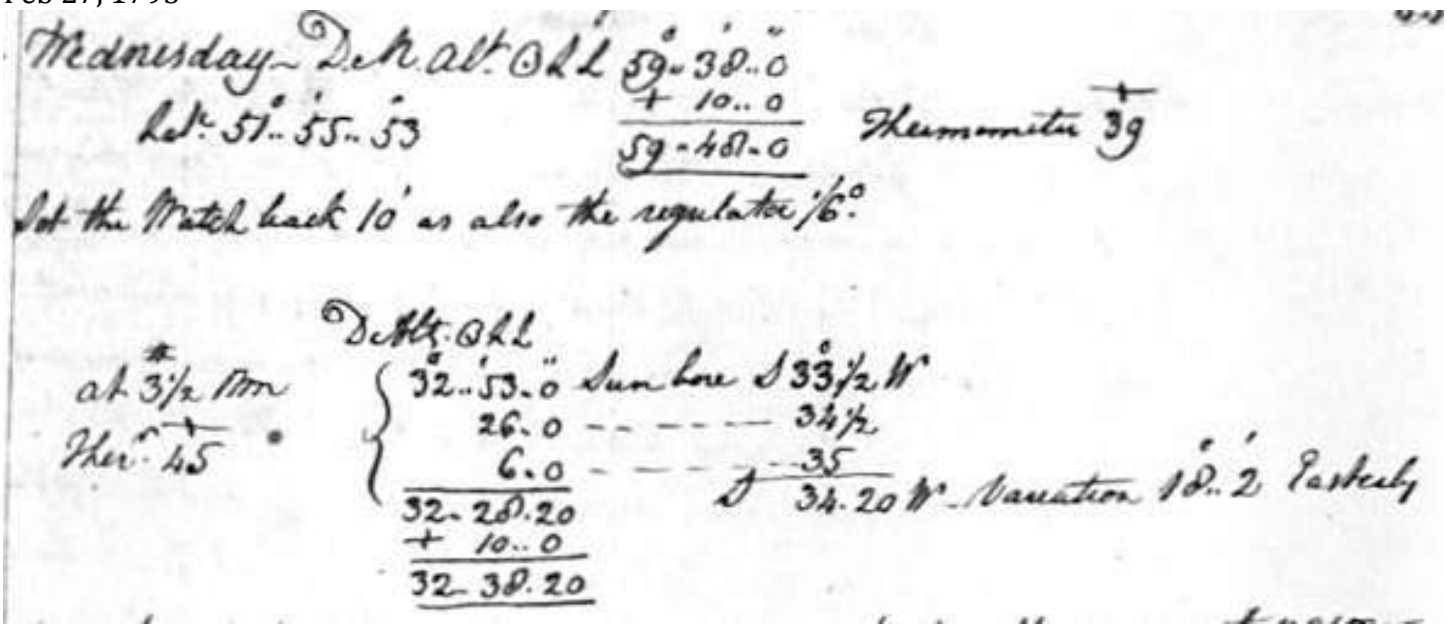
ECLIPSES of the SATELLITES of JUPITER.
MEAN TIME.

I. Satellite.		II. Satellite.		III. Satellite.	
Emersions.		*Emersions.*			
Days.	H. M. S.	Days.	H. M. S.	Days.	H. M. S.
2	14. 18. 3	1	14. 38. 56	1	5. 35. 20 Im.
* 4	8. 47. 0	5	3. 56. 31	* 1	9. 9. 40 E.
6	3. 15. 49	8	17. 14. 9	* 8	9. 36. 12 Im.
7	21. 44. 46	12	6. 31. 45	8	13. 10. 34 E.
9	16. 13. 35	15	19. 49. 24	15	13. 37. 10 Im.
*11	10. 42. 34	*19	9. 7. 4	15	17. 11. 34 E.
13	5. 11. 23	22	22. 24. 45	22	17. 38. 23 Im.
14	23. 40. 22	26	11. 42. 27	22	21. 12. 48 E.
16	18. 9. 13	30	1. 0. 9	29	21. 40. 30 Im.
18	12. 38. 12			30	1. 14. 54 E.
*20	7. 7. 3				
22	1. 36. 2				
23	20. 4. 54			IV. Satellite.	
25	14. 33. 54				
*27	9. 2. 46			16	16. 55. 14 Im.
29	3. 31. 48			16	21. 35. 28 E.
30	22. 0. 40				

O 2

On the 8th, we can see that moon II emerges from Jupiter at 17h14m9s GAT. Fidler knows that his longitude is approximately 92°W so he is a bit over 6 hours behind Greenwich time. He can start watching for the appearance of moon II just after 11:00 PM and he sees it at 11h17m54s per his pocket watch. He corrects for the difference between his watch and LAT from his most recent time sight (-10m12s) to get a local time for emersion of 11h7m42s. It is then just a matter of subtracting from Greenwich time and converting to degrees to find his longitude of 91°36'45". Fidler writes down 92°13 ¾', a difference of about 37'. Since it's just a simple subtraction, it's really not clear why there is a difference unless it's just a copying error. The difference isn't significant as the emersions of Jupiter's satellites could not be reliably timed to better than a minute using a hand held telescope of at least 40x magnification and the predictions weren't all that reliable until about 1830.

Variation of the Compass
Feb 27, 1793


Wednesday – Det. alt. ⊙ ☾ L 59°-38'-0"
+ 10'-0
59°-48'-0
Lat. 51°-55'-53
Thermometer 39
Set the Watch back 10' as also the regulator 1/6°

Det. Alt. ⊙ ☾ L
at 3½ PM
Ther. 45
32°-53'-0" Sun bore S 33½ W
26-0 ---------- 34½
6-0 ---------- 35
32-28-20
+ 10-0
32-38-20
S 34-20 W. Variation 1°-2' Easterly

Since all of Fidler's courses are taken using a magnetic compass, he needs to know the difference between true North and magnetic North throughout his travels in order to convert his courses to map coordinates. In 1800 the magnetic North pole was somewhere around King William Island in Northern Canada; this caused the variation between true and magnetic North to vary significantly over the area that Fidler traversed. He had to make compass measurements a regular part of his navigation routine in order to correct his courses. Here he takes the sun's altitude at 3:30PM on Feb. 27, 1793 in addition to a compass bearing for the sun's position.

Finding the azimuth of a body is straightforward from a solution of the celestial triangle in spherical trigonometry. The equation is:
Azimuth = arccos[(sin(altitude)*sin(latitude)-sin(declination))/(cos(altitude)*cos(latitude))]
So we need to know the latitude, declination, and altitude of the body. The declination comes out of the nautical almanac as does the sun's semidiameter needed for finding the true altitude of the sun's center. Looking into the February pages of the NA for 1973 we find:

[14] FEBRUARY 1793. II.

Days of the Month.	Days of the Week.	Sun's Longitude.	Sun's Right Asc. in Time.	Sun's Declin. South.	Equat. of Time. Add.	Diff.
		S. D. M. S.	H. M. S.	D. M. S.	M. S.	S.
1	F.	10. 13.12.12	21. 2.42,0	16.52.19	14. 7,4	6,7
2	Sa.	10. 14.13. 1	21. 6.45,2	16.34.50	14. 14,1	5,9
3	Su.	10. 15.13.49	21. 10.47,7	16.17. 5	14. 20,0	5,1
4	M.	10. 16.14.36	21. 14.49,4	15.59. 3	14. 25,1	4,3
5	Tu.	10. 17.15.22	21. 18.50,3	15.40.43	14. 29,4	3,6
26	Tu.	11. 8.25.32	22. 40.15,2	8.25. 8	13. 6,9	10,7
27	W.	11. 9.25.40	22. 44. 0,5	8. 2.34	12. 55,7	11,2
28	Th.	11. 10.25.47	22. 47.45,3	7.39.52	12. 43,9	11,8

From here we can proportion the sun's declination for 3:30 LAT using a longitude of account of 112°50' giving GAT of 11h1m and a resulting declination of 7°52'09" S. The latitude has been calculated at 51°55'53" and the double altitude is halved and then corrected for refraction and semidiameter to get 16°32'1" as the true altitude.

III. FEBRUARY 1793. [15]

Days of the Month.	Semidiameter of the Sun.	Time of D° passing the Meridian.	Hourly Motion of the Sun.	Logarithm of the Sun's Distance.	Place of the Moon's Node.
	M. S.	M. S.	M. S.		S. D. M.
1	16. 16,5	1. 8,0	2. 32,1	9,993846	5. 16. 55
7	16. 15,5	1. 7,4	2. 31,7	9,994324	5. 16. 36
13	16. 14,3	1. 6,7	2. 31,4	9,994844	5. 16. 17
19	16. 13,0	1. 6,1	2. 31,0	9,995401	5. 15. 58
25	16. 11,6	1. 5,6	2. 30,5	9,996009	5. 15. 39

The recipe for finding the true azimuth using log tables is then:

Add the complement of the latitude,
The complement of the altitude,
And the sun or star's polar distance into one sum.
From half this sum, subtract the polar distance and note the half sum and the remainder.
Add together
The log cosecant of the comp. latitude, rejecting the index
The log cosecant of the comp altitude, rejecting the index
The log sine of the half-sum
And the log sine of the remainder, into one sum;
Half the sum of these four logs will be the log cosine of the half of true azimuth.

	deg	min	sec			
90-lat	38	4	6	cosec	0.20999	
90-alt	73	27	58	cosec	0.01834	
90+dec	97	52	9			
sum	209	24	15			
half sum	104	42	7	log sin	9.98554	
PD	97	52	9			
remainder	6	49	57	log sin	9.07544	
				sum	19.28932	
				1/2 sum	9.64466	
				acos	63.81807	
				double	127.63614	
					52.36386	W of S

When compared with Fidler's measured azimuth of 34°20', this gives a variation of only 18°2', exactly the same as Fidler's value.